Better Judges, Same Answers: When Self-Verification Can and Cannot Improve Mathematical Reasoning

Aryama V Murthy
Mukund Hebbar
Yashas Nataraj Kotre
Anivarth Varma Penumetcha
IIIT Hyderabad

Abstract

Training a 2B model to judge its own mathematical attempts raised its AUROC from 0.57 to 0.67, yet picking among its answers did not improve. We show this is predicted, not anomalous. Three measurement illusions hide what a self-verifier can do: (i) a lenient judge looks near chance under sampled verdicts even when its underlying score discriminates; (ii) on a typical pool, 99.9% of the pairs behind an AUROC compare different questions, while selection and repair provably use only within-question information; and (iii) majority vote saturates, and before that point verification cannot beat it at equal compute.

A five-parameter model fitted only to 4-attempt aggregates predicts the 16-attempt scaling curves of 16,000 new attempts to within a point. Reading verdict probabilities instead of sampling verdicts raises the judge's AUROC by 0.07 to 0.09 (paired 95% CIs exclude zero). Judge-free trace features match the trained judge's pooled AUROC yet are at chance on contested questions, exactly as the theory requires.

The theory predicts a procedure, contest-adaptive verification, that spends judgement only where sampled answers disagree. In a pre-registered test on 800 questions it beats majority vote at equal compute by $+1.6$ points $[+0.5, +2.6]$ and passes the plateau that voting and adaptive-consistency share, close to the $\approx +1.3$ the model predicted before judging.

1 Introduction

A model that can check its own work should be able to pick its best answer. We trained a 2B-parameter model, Qwen3.5-2B, to judge its own mathematical attempts with answer-checker labels. Its AUROC on its own attempts rose from 0.57 to 0.67, and the gain replicated on a held-out test set. Yet picking one of four answers with the trained judge was no better than majority vote (75.2% either way), and flagging and regenerating attempts recovered 0.4 of the 7 points that perfect flags would have recovered. Every off-the-shelf judge we tried looked "near chance" on the model's own attempts.

We show that these outcomes are what the mathematics predicts. They come from three measurement illusions:

1. **Leniency.** A judge that almost always says "correct" has a near-chance AUROC under sampled verdicts, even when its underlying score discriminates (§4.1).

2. **Pooled AUROC.** On a typical evaluation pool, 99.9% of the pairs behind an AUROC compare *different* questions. Picking and repair use only within-question information, exactly (Theorem 1, §4.2).

3. **The compute regime.** Majority vote saturates. Verification cannot beat majority vote at equal compute before that point, and can after it (§5).

A model with five parameters reproduces the measured numbers, including three it was not fitted to (Table 1). It also predicts a simple procedure, *contest-adaptive verification* (CAV), and a training change (per-question balance), both of which we test on the model's own attempts (§7).

Contributions.

- A calibrated model of self-verified answer selection.
- An invariance theorem with an exact repair identity, a closed form for the gain from tie-breaking, and a saturation-and-consistency account of when verification pays.
- Measurements of judge leniency and within-question discrimination on 16 own attempts per question.

- CAV, evaluated against majority vote and adaptive-consistency at equal compute.
- A proposition and an experiment on training judges for selection rather than for verdict accuracy.

2 Setting

Models and data. The solver and the judge are the same model: Qwen3.5-2B with a LoRA adapter trained by Dr. GRPO (Liu et al., 2025) on equal thirds of three tasks: solving, judging injected errors, and reconstructing questions. Questions come from DeepMath-103K (He et al., 2026): the 400-question test split, a 200-question development slice, and 400 held-out training questions that the adapter never saw.

Protocol. For each question we sample 16 attempts (temperature 1.0, 16k-token cap). Answers are graded by a rule-based checker. The judge sees an attempt split into numbered steps and writes a verdict after optional reasoning (8k cap).

We read two scores:

- the probability of the verdict token after a reasoning chain;
- a *direct* probability, read right after the prompt with no reasoning (Kadavath et al., 2022).

Selection rules are averaged over 100 random orders of each question's attempts. Differences are bootstrapped over questions (Miller, 2024). The protocol was pre-registered before any attempt was generated (`phase3/PREREGISTRATION.md`).

3 A Model of Self-Verified Answer Selection

Solver. For a question x the solver draws attempts $y_1, \dots, y_N$ independently. Each attempt is right ($Z_i = 1$) with probability p_x, and right attempts share the true answer. A wrong attempt gives the question's *attractor* wrong answer with probability ρ, and otherwise a wrong answer of its own. Solve rates are heterogeneous: $p_x \sim \mathrm{Beta}(a, b)$. Majority vote picks the most common answer and breaks ties uniformly.

Judge. The judge sees one attempt at a time. It has a latent score

$$u_i = c_0 + \gamma z_x + d\, Z_i + \eta_i, \qquad \eta_i \sim \mathcal{N}(0,1), \quad (1)$$

Quantity (not used in fitting)	Model	Measured
Always right in 8 tries	0.43	0.39
Always wrong in 8 tries	0.12	0.11
Repair gain, mixed-judge flags (pts)	+0.30	+0.38
Judge AUROC, 4 sampled verdicts	0.568	0.568
Dev majority@8 / @16 (%)	81.6 / 82.5	80.2 / 81.3
Dev pass@16 (%)	94.1	92.0
Test majority@8 / @16 (%)	77.5 / 78.2	77.5 / 78.7
Test pass@16 (%)	90.8	90.2

Table 1: Out-of-sample checks of the model, fitted only to the Phase-2 4-attempt aggregates. Rows 1–3 use the Phase-2 report. Row 4 uses a judge fitted only to single-verdict flag rates (§4.1). Rows 5–8 use the 16-attempt pools of this paper.

and each sampled verdict says "correct" with probability $\Phi(u_i)$. Here z_x is the standardised logit of p_x, so γ is a *between-question* (difficulty) signal, d is the *within-question* signal, and c_0 is the judge's leniency. Optionally η is shared, with weight ρ_s, by attempts that give the same answer. With V sampled verdicts, q_i is the share of "correct" verdicts.

Calibration. Three numbers per split fit (a, b, ρ) exactly: single-attempt accuracy, pass@4 and majority@4. On the test split $(67.4, 82.8, 74.0)$ this gives $\mathrm{Beta}(0.46, 0.22)$ and $\rho = 0.37$. The fit is U-shaped: questions are mostly almost-always-right or almost-always-wrong.

The model is checked on numbers that were not used to fit it (Table 1):

- the share of training questions answered right (or wrong) in all of 8 tries;
- the accuracy gain from regenerating the attempts the mixed judge flagged;
- the AUROC of the untrained judge with four sampled verdicts (§4.1);
- majority vote and pass@N at 16 attempts per question.

4 Three Measurement Illusions

4.1 Leniency hides discrimination

A binary verdict with hit rate H (the share of wrong attempts it flags) and false-alarm rate F has

$$\mathrm{AUROC} = \tfrac{1}{2}\,(1 + H - F), \qquad (2)$$

which is close to $\frac{1}{2}$ whenever the judge rarely says "incorrect", whatever its discrimination. In Phase 2

the mixed judge flagged 7.4% of wrong own attempts and 2.2% of right ones. That gives an AUROC of 0.526 (reported as 0.53, "near chance"). In signal-detection terms it is $d' = z(H) - z(F) = 0.57$.

Fitting Eq. 1 to these two rates gives $c_0 = 2.05$ and $d = 0.80$. With no further fitting the same judge predicts AUROC 0.568 with four sampled verdicts, the value later measured for that judge on test attempts. Its latent score has AUROC 0.715.

Moving only c_0, to about 28% "incorrect" verdicts, raises the four-verdict AUROC to 0.665, the value reached by self-judge training on a 49%-wrong training set. The training "gain" is therefore reproduced by a criterion shift, with discrimination unchanged. Section 7.1 tests this account directly.

4.2 Pooled AUROC is between-question AUROC

Split the pairs behind a pooled AUROC into same-question and cross-question pairs. With k_x right attempts out of N on question x, and $K = \sum_x k_x$ over Q questions,

$$A_{\text{pooled}} = wA_{\text{within}} + (1-w)A_{\text{between}},$$
$$w = \frac{\sum_x k_x(N-k_x)}{K(QN-K)}. \tag{3}$$

In expectation $w \approx \frac{N-1}{NQ} \cdot \frac{\mathbb{E}[p(1-p)]}{\bar{p}(1-\bar{p})}$, which falls like $1/Q$. On a 400×4 pool, $w = 0.076\%$, so the pooled AUROC is a between-question AUROC. A score that knows only p_x, and nothing about the attempt, reaches pooled AUROC 0.94. Bulut (2026) report the same confound for hidden-state probes.

The next result says why this matters.

Theorem 1 (invariance). Let $f_i \in [0,1]$ flag attempt i for resampling. The expected change in accuracy is

$$\Delta = \mathbb{E}[f\,(p_x - Z)] = -\mathbb{E}_x\big[\mathrm{Cov}(f, Z \mid x)\big]. \tag{4}$$

For selection, the Bayes-optimal choice among a question's answers depends on the verifier only through the within-question likelihood ratios $\Lambda_x(s) = f(s \mid x, Z{=}1)/f(s \mid x, Z{=}0)$.

Proof. Appendix A.

Consequences.

- Any flag or score that depends only on the question has $\Delta = 0$ exactly and reduces selection to majority vote, whatever its pooled AUROC.

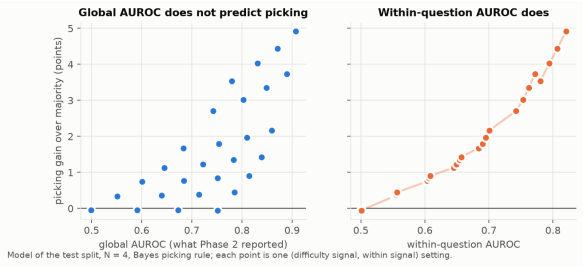


Figure 1: Picking gain over majority vote in the calibrated model ($N = 4$, Bayes rule). Left: against pooled AUROC. Right: against within-question AUROC.

- Random flags have $\Delta = 0$ (Phase 2: $69.75 \to 69.75$).
- The optimal flag is question-relative: flag iff $P(Z{=}1 \mid x, s) < p_x$.

Figure 1 shows the point in the model. Raising the difficulty signal moves pooled AUROC from 0.50 to 0.75 with no gain in picking, while picking accuracy is a single curve in the within-question AUROC.

4.3 Most of the headroom is ties

At $N = 4$ the 8.7-point gap between majority vote and pass@4 splits as follows:

- **Ties (5.7 points):** configurations such as (right, wrong, wrong, wrong) all different, where majority vote is a coin flip.
- **Overturns (3.0 points):** the right answer has to beat a larger wrong group.

Proposition 1 (tie-breaking). Take majority vote and let the judge decide only among tied groups. If right scores are $\mathcal{N}(d,1)$ and wrong scores $\mathcal{N}(0,1)$, independently, then

$$\text{gain} = P_{\text{1-1-1-1}}\Big(M_4(d) - \tfrac{1}{4}\Big) + P_{\text{2-2}}\Big(\Phi(d) - \tfrac{1}{2}\Big) \geq 0, \tag{5}$$

where $M_4(d) = \int \phi(t-d)\Phi(t)^3\,dt$ and $P_{\cdot}$ are the configuration shares (6.7% and 1.3% on test). The closed form matches simulation to within 0.07 points.

By contrast, Phase 2's summed-confidence rule can fall below majority vote: a single confident wrong attempt can outscore two right ones.

Overturning a plurality of m with a single right attempt under the Bayes rule (vote weight β) succeeds with probability $\Phi\big([(1-m)\beta + (1+$

$m)d^2/2]/(\sqrt{1+m}\,d)\big)$, which is small for weak judges.

The headroom also sits on hard questions: mean solve rate 0.33, against 0.54 for the pairs a within-question AUROC averages over. We therefore report within-question AUROC on *contested* questions, where majority vote does not already pick the right answer.

5 When Verification Pays

Majority vote saturates. As $N \to \infty$, majority vote is right on question x if and only if p_x exceeds the mass of every wrong answer; in the model, iff $p_x > \rho(1-p_x)$. That probability is 78.3% on the test calibration, while pass@$N \to 100\%$. Simulated majority accuracy reaches 77.7, 78.2 and 78.8 at $N = 8$, 16 and 64, a plateau also reported for large-sample voting (Brown et al., 2024).

Proposition 2 (aggregation is consistent, argmax is not). Score each answer group C_a by $\sum_{i \in C_a} \log \Lambda(s_i)$, and suppose judge errors are conditionally independent. Then the score divided by N converges to

$$\pi(a)\,\mathrm{KL}(f_1 \| f_0) > 0 \text{ for the right answer,}$$
$$-\pi(a)\,\mathrm{KL}(f_0 \| f_1) < 0 \text{ for a wrong one,}$$

so accuracy tends to $P(p_x > 0)$ for *any* judge with $d > 0$. Picking the single top-scored attempt instead converges to the precision of the top score bin, the imperfect-verifier ceiling of Stroebl et al. (2026).

If judge errors are shared within an answer group (correlation ρ_s), a right-versus-wrong group comparison is capped at $\Phi(d/\sqrt{2\rho_s})$.

The compute crossover. Charge an attempt 1 unit and a judge verdict 0.3. In the model, verification cannot beat majority vote at equal compute for budgets of 8 units or fewer, even with within-question AUROC 0.78. It wins only after majority vote saturates: +0.6 to +3.0 points at 16 units and +1.9 to +6.1 at 32. Phase 2 tested picking at 4 attempts with 4 verdicts each, inside the region where verification must lose.

Contest-adaptive verification (CAV). These results suggest a simple procedure:

1. Sample 4 attempts. If all four give the same answer, stop: about 50% of questions, right 99.4% of the time in the model.
2. Otherwise sample up to N_1 attempts.
3. Judge only the attempts in the two largest answer groups, at most m per group.
4. Pick the group maximising $n_g\,(\beta + \overline{\mathrm{logit}\,P(\text{correct})})$, where n_g is the group size and β is a vote weight tuned on development data.

Adaptive stopping alone (Aggarwal et al., 2023) reaches the majority plateau with less compute, but cannot pass it; only the judge can. In the model CAV gains +1.7 to +5.3 points over majority vote at equal average cost (≈ 11 units). The gain holds when attractor strength varies across questions, when the judge is weaker on hard questions, and when judge errors are shared within answer groups (Appendix B). Section 7.2 tests CAV on the model's own attempts.

6 Training a Judge for Selection

Outcome-labelled verdict training rewards a judge for predicting whether an attempt's answer is right.

On our pools, 56% of the label entropy is question difficulty:

$$H(Z) = 0.91 \text{ bits} = \underbrace{I(Z;X)}_{0.51} + \underbrace{\mathbb{E}[H(p_x)]}_{0.40}.$$

With 4 attempts per question, 53% of wrong attempts come from questions with no right attempt. These carry no within-question contrast. On a globally balanced set like Phase 2's, a judge that knows only p_x earns 86% verdict reward.

Proposition 3 (pointwise–selection misalignment). Suppose the judge sees one attempt with a difficulty-like cue $x_b = s_b z_x + e_b$ that varies between attempts of the same question (length, for example), and a within cue $x_w = dZ + e_w$. For a score $\theta_b x_b + \theta_w x_w$,

$$d'_{\text{within}} = d\,\theta_w / \sqrt{\theta_b^2 + \theta_w^2}. \tag{6}$$

- When $P(Z \mid x)$ varies with x, the pointwise-optimal θ_b is positive and strictly lowers within-question discrimination.
- Class rebalancing changes only the intercept.
- Under *per-question balance* ($P(Z{=}1 \mid x) = \frac{1}{2}$ for every x), x_b carries no information and $\theta_b = 0$. The initial gradient on any question-only feature is then $\mathbb{E}[(\frac{1}{2} - Z)\varphi(x)] = 0$.

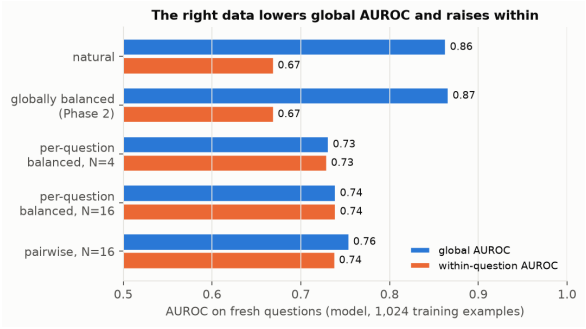


Figure 2: Learning model (1,024 examples from 900 questions). Per-question balanced and pairwise data trade pooled AUROC for within-question AUROC.

In a finite-sample learning model with 64-dimensional features, 900 questions and 1,024 training examples, per-question balanced or pairwise data raise within-question AUROC from 0.67 to 0.74 while *lowering* pooled AUROC from 0.86 to 0.74 (Figure 2). Judge checkpoints must therefore be compared on within-question AUROC, not pooled AUROC.

Per-question pairs scale as $N(N-1)\,\mathbb{E}[p(1-p)]$ per question: 1.07 at $N = 4$ and 21.4 at $N = 16$.

7 Experiments

We generate 16 attempts for each of 1,000 questions (dev 200, test 400, extra 400) with the mixed adapter, and judge every answered attempt with the same adapter. All settings are tuned on dev and evaluated on test ∪ extra.

The pipeline reproduces Phase 2, and the model predicts the scaling curves. The Phase-3 pools match the Phase-2 numbers:

- **Dev:** single attempt 69.8%, pass@4 86.3%, majority@4 78.1% (Phase 2: 69.8, 86.5, 77.5).
- **Test:** 67.0, 82.8, 74.9 (Phase 2: 67.4, 82.8, 74.0).

The model was fitted only to the Phase-2 4-attempt aggregates. On test it predicts majority@8 = 77.5 (observed 77.5), majority@16 = 78.2 (observed 78.7) and pass@16 = 90.8 (observed 90.2). It also predicts the share of 4-attempt configurations where the majority is already right (71.7% vs 72.3%) and where no attempt is right (17.2% vs 17.3%).

Majority vote saturates at the predicted limit (78.3%) while pass@16 reaches 90.2% (Figure 5). On test, adaptive-consistency reaches the same plateau (78.7%) with fewer attempts but cannot pass it.

Judge-free features already reach the trained judge's pooled AUROC. Table 2 scores attempts by length, by counts of self-correction phrases, or by guess phrases.

- **Pooled:** these features reach AUROC 0.65–0.70, the level the Phase-2 trained self-judge reached (0.665). Hard questions produce long, hesitant attempts.
- **Within question:** 0.54–0.61.
- **Contested questions:** 0.46–0.58, about chance. This is where picking happens (§4.3).

The table is the invariance theorem in miniature: a large pooled AUROC that selection cannot use.

The same data support the liveness account (Appendix A): 98–99% of *right* attempts contain self-corrections, about 25 per attempt (36 per wrong attempt). A detector of "an error occurred somewhere" therefore fires on almost every trace. What decides correctness is whether an error is still live at the end.

7.1 Stage 0: what the judge knows

Table 3 tests the leniency account of §4.1 on test attempts 0–3, judged with four chains each.

- **The judge is lenient.** It says "incorrect" to about one attempt in seven; $d' = 0.49$.
- **Sampled verdicts look near chance.** One verdict gives AUROC 0.57; the share of four gives 0.64.
- **Probabilities recover the signal.** The verdict probability averaged over the four chains, and the direct probability read with no reasoning at all, reach 0.69–0.71. That is the latent AUROC the model inferred from Phase-2 flag rates alone (0.715). Both paired differences from the four-verdict share are significant: +0.067 [+0.047, +0.085] and +0.090 [+0.049, +0.130] (paired bootstrap over questions).
- **Verdicts on one attempt are strongly correlated.** Four "incorrect" verdicts on the same wrong attempt occur about 12× as often as independent verdicts would allow. This is the persistent per-attempt component of Eq. 1.

Set	Trace feature	Pooled AUROC	Within-question	Contested within	Mean (right / wrong)
dev	Length (tokens)	0.700	0.580	0.495	7521.8 / 10005.3
dev	Self-corrections (“wait”, “actually”, . . .)	0.677	0.563	0.477	23.8 / 34.9
dev	Guess phrases (“let’s assume”, . . .)	0.680	0.611	0.577	1.0 / 2.4
dev	Final-answer tags	0.529	0.549	0.488	1.0 / 1.2
test	Length (tokens)	0.655	0.573	0.486	7977.9 / 9750.5
test	Self-corrections (“wait”, “actually”, . . .)	0.648	0.553	0.514	26.0 / 36.0
test	Guess phrases (“let’s assume”, . . .)	0.656	0.576	0.568	1.1 / 2.3
test	Final-answer tags	0.526	0.536	0.509	1.0 / 1.1

Table 2: Judge-free trace features. They reach the pooled AUROC of the Phase-2 trained self-judge (0.665) because hard questions produce long, hesitant attempts, but are near chance on contested questions, where picking happens. Almost every right attempt contains self-corrections.

Test attempts 0–3, 4 chains each	Measured	Model
Share of “incorrect” verdicts (sampled)	0.154	0.04 (Phase 2 dev)
Hit rate H / false-alarm rate F	0.255 / 0.125	0.074 / 0.022
$d' = z(H) - z(F)$	0.49	0.57
AUROC, 1 sampled verdict	0.565	0.526
AUROC, share of 4 sampled verdicts	0.635	0.568
AUROC, verdict probability, 1 chain	0.689	–
AUROC, verdict probability, mean of 4 chains	0.694	$\leq$0.715
AUROC, direct probability (no reasoning)	0.717	–
P(4 of 4 “incorrect” | wrong), observed	0.0422	0.005
same, if verdicts were independent	0.00352	0.00003
Chains reaching a verdict	0.902	0.89 (Phase 2)

Table 3: Stage 0: the untrained mixed judge on its own test attempts (1486 answered attempts, 28.7% wrong). “Model” is the latent judge fitted to Phase-2 single-verdict flag rates (§4.1).

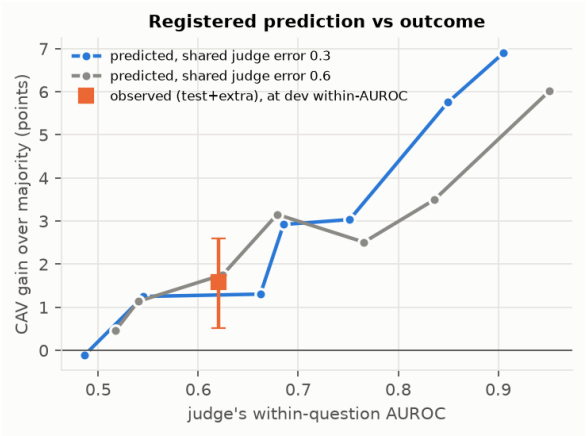


Figure 3: CAV gain as a function of the judge’s within-question AUROC, predicted before any judging from the real dev answer structure with simulated judges (lines). The square is the observed pre-registered result on test $\cup$ extra (95% CI), placed at the real judge’s dev within-question AUROC.

Within questions (Table 4), the judge’s discrimination is lower than pooled, and lowest on contested questions, where the headroom lies. The verification signal is weakest where selection needs it: the configuration analysis of §4.3 anticipated this.

7.2 Stage 1: contest-adaptive verification

Vote-only methods plateau. On test $\cup$ extra (800 questions), majority vote rises from 76.0% at $N = 4$ to 79.8% at $N = 16$, while pass@16 is 91.4%. Adaptive-consistency reaches 79.8% with 10.4 attempts on average, and no threshold takes it higher (Table 6).

CAV passes the plateau (pre-registered primary test). CAV uses the chain probability with $N_1 = 16$, $m = 8$ and β tuned on dev. It reaches 81.0% at an average cost of 11.3 units (114k decoded tokens per question). Against majority vote at the same cost (79.4%) the difference is $+1.6$ points $[+0.5, +2.6]$. At matched decoded tokens it is $+1.4$ $[+0.4, +2.5]$.

Against adaptive-consistency at the closest cost the difference is $+1.2$ $[+0.05, +2.3]$. CAV is also above majority@16 (79.8%) while using 29% fewer units.

The size of the gain was predicted before any judging. Feeding the judge’s dev within-question AUROC (0.62–0.63) into the semi-synthetic model gave $\approx +1.3$ points.

Where does the gain come from? Classify each question by its 16-attempt answer distribution.

- **Right answer in the minority** (10.6% of questions): CAV gains $+11.1$ points on average, $+1.18$ of the total $+1.60$.
- **Ties** (1.8%): $+0.28$.
- **Majority already right** (79%): $+0.14$.
- **No right attempt** (8.6%): exactly 0, as Theorem 1 requires.

At $N = 16$, verification wins mainly by overturning wrong pluralities, not by breaking ties.

Set	Single attempt	Score	Pooled AUROC	Within-question AUROC	Contested within-question AUROC
dev	0.698	chain	0.692	0.620 [0.578, 0.660]	0.612 [0.508, 0.712]
		direct	0.730	0.634 [0.576, 0.692]	0.596 [0.460, 0.732]
		length	0.699	0.580 [0.537, 0.623]	–
test	0.670	chain	0.682	0.611 [0.573, 0.645]	0.511 [0.423, 0.604]
		direct	0.718	0.614 [0.566, 0.658]	0.469 [0.375, 0.575]
		length	0.655	0.573 [0.541, 0.606]	–
extra	0.682	chain	0.713	0.668 [0.638, 0.700]	0.631 [0.561, 0.703]
		direct	0.746	0.661 [0.622, 0.699]	0.641 [0.551, 0.731]
		length	0.688	0.596 [0.568, 0.627]	–

Table 4: Judging own attempts (16 per question; first chain). "Length" scores an attempt by its token count. Brackets: 95% question-cluster bootstrap intervals.

	Majority right	Tie	Overturn	Unwinnable
dev	0.749	0.066	0.050	0.135
test	0.723	0.062	0.042	0.173
extra	0.743	0.075	0.043	0.139
model (test calibration)	0.717	0.080	0.030	0.172

Table 5: Shares of 4-attempt configurations (random 4-subsets of the 16 attempts). Only ties and overturns can be won by a judge.

Cheap verification pays even at small budgets. CAV with the direct probability needs only a prefill pass per judged attempt. With $N_1 = 8$, $m = 4$ it reaches 79.2% at 6.1 units, $+1.3$ $[+0.8, +1.9]$ over majority vote at the same cost ($+1.3$ $[+0.8, +1.8]$ at matched tokens). Adaptive-consistency is as good at that budget (-0.5 $[-1.2, +0.2]$).

The crossover of §5 assumed a verdict costs 0.3 of an attempt. With prefill-only reads, it moves to smaller budgets, as the cost term predicts.

Argmax is worse than voting; aggregation is not. Picking the single attempt with the best verdict probability gives 76.7% at $N = 16$, *below* majority vote (79.8%). Summing log-odds over answer groups gives 80.9%, as Proposition 2 predicts. CAV gets the aggregation gain at about half the cost by judging only contested questions.

A learned combination raises AUROC but not picking. The exploratory combiner (fitted on dev) raises within-question AUROC on extra to 0.717 (test 0.666). It does not improve CAV accuracy (Table 6). Within-question AUROC averaged over all questions is the wrong target: the gain has to come from contested questions, where the combiner is still near chance on test.

Training the judge (future work). We also retrained the judge with the Phase-2 recipe in two arms: globally balanced, as in Phase 2, and per-question balanced (Proposition 3). Verdict reward rose from 40% to 57% and from 48% to 53% respectively; the smaller rise is consistent with the removed difficulty shortcut. We did not evaluate the trained judges on the selection pools, so Proposition 3 rests on theory and the learning model of §6.

Rule	Acc. (%)	Cost	Tokens
Majority, $N = 4$	76.0	4.00	35.7k
Majority, $N = 8$	78.7	8.00	71.4k
Majority, $N = 12$	79.5	12.00	107.1k
Majority, $N = 16$	79.8	16.00	142.8k
pass@16 (oracle)	91.4	16.00	142.8k
Adaptive-consistency, $t = 0.8$	79.1	4.01	40.5k
Adaptive-consistency, $t = 0.9$	79.7	6.41	64.0k
Adaptive-consistency, $t = 0.99$	79.8	10.39	99.6k
Bayes aggregation, $N = 4$	76.7	5.12	47.5k
Bayes aggregation, $N = 16$	80.9	20.49	189.7k
Best-of-N argmax, $N = 4$	75.3	5.12	47.5k
Best-of-N argmax, $N = 16$	76.7	20.49	189.7k
CAV (chain), $N_1 = 8, m = 4$	79.0	6.70	65.2k
CAV (direct), $N_1 = 8, m = 4$	79.2	6.09	56.1k
CAV (combo-cheap), $N_1 = 8, m = 4$	79.0	6.09	56.1k
CAV (direct), $N_1 = 12, m = 4$	79.7	8.08	76.5k
CAV (chain), $N_1 = 12, m = 8$	80.3	9.15	91.1k
CAV (direct), $N_1 = 16, m = 4$	80.1	10.06	96.8k
CAV (chain), $N_1 = 16, m = 8^\star$	81.0	11.31	113.9k
CAV (combo), $N_1 = 16, m = 8$	79.6	11.31	113.9k

Table 6: Selection on test ∪ extra (800 questions), settings tuned on dev. Cost: attempt $= 1$, judge chain $= 0.3$, direct read $= 0.05$; tokens: decoded tokens per question. *: pre-registered primary rule.

8 Related Work

Verifiers and selection. Outcome verifiers for best-of-N (Cobbe et al., 2021), process supervision (Lightman et al., 2023) and step-level benchmarks (Zheng et al., 2025) measure verifier quality per solution. We instead measure what selection and repair can use.

Weighted voting with log-odds weights is classical (Nitzan and Paroush, 1982; Berend and Kontorovich, 2014). Combinations of weak verifiers (Saad-Falcon et al., 2025) and resampling limits under imperfect verifiers (Stroebl et al., 2026) bound

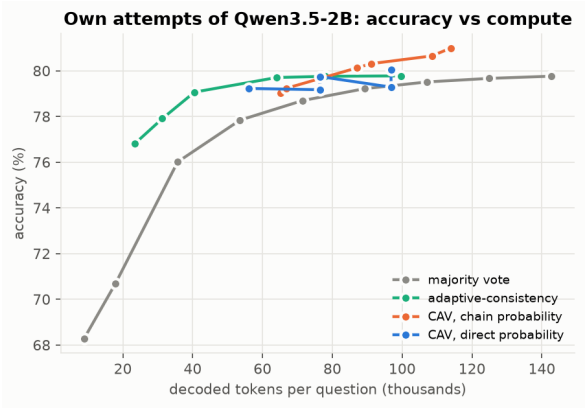


Figure 4: Accuracy against decoded tokens per question on test ∪ extra (own attempts of Qwen3.5-2B). Settings tuned on dev.

what verification can add. Compute-optimal test-time scaling (Snell et al., 2024), adaptive stopping for self-consistency (Wang et al., 2023; Aggarwal et al., 2023) and scaling models of voting systems (Chen et al., 2024; Brown et al., 2024) motivate our compute-matched comparisons.

Explaining aggregate behaviour with per-problem heterogeneity. Schaeffer et al. (2025) explain power-law scaling of repeated sampling with a heavy-tailed distribution of per-problem success rates. Our Beta-distributed solve rates play the same role for verification.

Self-verification and its limits. Self-correction without external feedback often fails (Huang et al., 2024; Stechly et al., 2025). Song et al. (2025) formalise a generation–verification gap that governs self-improvement. Our governing quantity is within-question discrimination on contested questions.

Bulut (2026) show that pooled probe AUROCs can come entirely from difficulty. We extend this to verifiers used for selection and repair, where it becomes an invariance theorem. Signal-detection analyses of LLM judges separate sensitivity from criterion (Cacioli, 2026). We show that verdict sampling turns a lenient criterion into a near-chance AUROC.

9 Limitations and Conclusion

Limitations.

- We study one solver family at 2B parameters and one judge (the same adapter).
- Labels come from an answer checker, so an attempt with flawed reasoning and a right answer counts as right.
- The step conversion of attempts is our own: Phase 2's converter was lost.
- The CAV gain is modest (+1.6 points) because the judge's discrimination on contested questions is weak.
- The length and liveness mechanisms (Appendix A) predict judging degrades with length and self-corrections. Within the 4k–16k-token range of our attempts we see no such degradation; only the premise that almost every right trace contains self-corrections is confirmed.
- Costs are counted in units and decoded tokens, not wall-clock time.

Conclusion. How good a self-verifier looks depends on what is measured.

- Sampled verdicts hide a lenient judge's signal: probabilities recover 0.07–0.09 AUROC.
- Pooled AUROC rewards knowing which questions are hard. Trace features alone match a trained judge pooled and are at chance where picking happens.
- Small-N picking puts verification where it cannot win.

A small model of these effects, fitted to 4-attempt aggregates, predicts 16-attempt behaviour to within a point. Measured within questions, scored by probability, and spent only on contested questions after self-consistency saturates, a 2B model's judgement of its own work beats compute-matched majority vote and adaptive-consistency.

References

Pranjal Aggarwal, Aman Madaan, Yiming Yang, and Mausam. 2023. Let's sample step by step: Adaptive-consistency for efficient reasoning and coding with LLMs. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*.

Daniel Berend and Aryeh Kontorovich. 2014. Consistency of weighted majority votes. *arXiv preprint arXiv:1312.0451*.

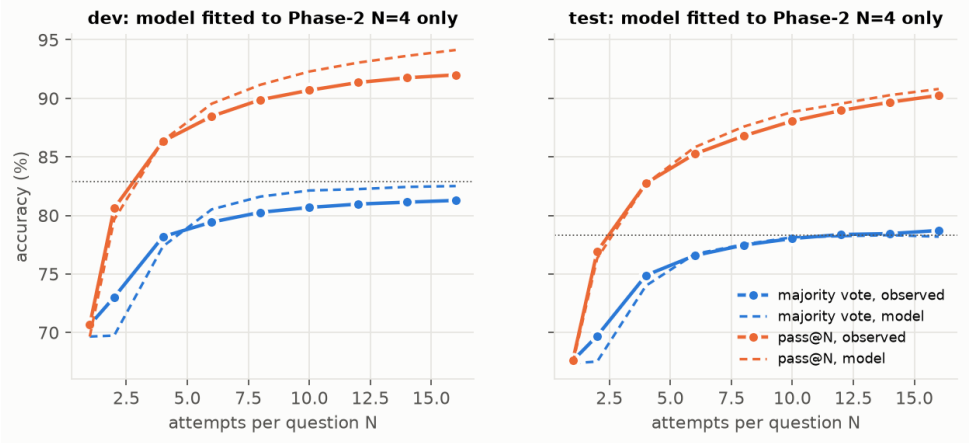


Figure 5: Majority@N and pass@N on own attempts (points) against the model fitted only to Phase-2 4-attempt aggregates (dashed). Dotted line: the predicted majority limit $P(p_x > \rho(1 - p_x))$.

Bradley Brown, Jordan Juravsky, Ryan Ehrlich, Ronald Clark, Quoc V. Le, Christopher Ré, and Azalia Mirhoseini. 2024. Large language monkeys: Scaling inference compute with repeated sampling. *arXiv preprint arXiv:2407.21787*.

Yigit Utku Bulut. 2026. It's the problem, not the path: Budget and difficulty confounds in LLM reasoning trajectories. *arXiv preprint arXiv:2609.03436*.

Cacioli. 2026. LLMs as signal detectors: Sensitivity, bias, and the temperature-criterion analogy. *arXiv preprint arXiv:2603.14893*.

Lingjiao Chen, Jared Quincy Davis, Boris Hanin, Peter Bailis, Ion Stoica, Matei Zaharia, and James Zou. 2024. Are more LLM calls all you need? towards the scaling properties of compound AI systems. In *Advances in Neural Information Processing Systems*.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*.

Zhiwei He, Tian Liang, Jiahao Xu, Qiuzhi Liu, Xingyu Chen, Yue Wang, Linfeng Song, Dian Yu, Zhenwen Liang, Wenxuan Wang, Zhuosheng Zhang, Rui Wang, Zhaopeng Tu, Haitao Mi, and Dong Yu. 2026. DeepMath-103K: A large-scale, challenging, decontaminated, and verifiable mathematical dataset for advancing reasoning. In *The Fourteenth International Conference on Learning Representations*.

Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. 2024. Large language models cannot self-correct reasoning yet. In *International Conference on Learning Representations*.

Saurav Kadavath et al. 2022. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*.

Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. 2023. Let's verify step by step. *arXiv preprint arXiv:2305.20050*.

Zichen Liu et al. 2025. Understanding R1-Zero-like training: A critical perspective. *arXiv preprint arXiv:2503.20783*.

Evan Miller. 2024. Adding error bars to evals: A statistical approach to language model evaluations. *arXiv preprint arXiv:2411.00640*.

Shmuel Nitzan and Jacob Paroush. 1982. Optimal decisive rules in uncertain dichotomous choice situations. *International Economic Review*, 23(2):289–297.

Jon Saad-Falcon et al. 2025. Weaver: Shrinking the generation-verification gap with weak verifiers. *arXiv preprint*.

Rylan Schaeffer et al. 2025. How do large language monkeys get their power (laws)? In *International Conference on Machine Learning*.

Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. 2024. Scaling LLM test-time compute optimally can be more effective than scaling model parameters. *arXiv preprint arXiv:2408.03314*.

Yuda Song, Hanlin Zhang, Carson Eisenach, Sham Kakade, Dean Foster, and Udaya Ghai. 2025. Mind the gap: Examining the self-improvement capabilities of large language models. In *International Conference on Learning Representations*.

Kaya Stechly, Karthik Valmeekam, and Subbarao Kambhampati. 2025. On the self-verification limitations of large language models on reasoning and planning tasks. In *International Conference on Learning Representations*.

Benedikt Stroebl, Sayash Kapoor, and Arvind Narayanan. 2026. The limits of inference scaling

through resampling. In *International Conference on Learning Representations*.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations*.

Chujie Zheng, Zhenru Zhang, Beichen Zhang, Runji Lin, Keming Lu, Bowen Yu, Dayiheng Liu, Jingren Zhou, and Junyang Lin. 2025. ProcessBench: Identifying process errors in mathematical reasoning. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*.

A Proofs

Pair-share identity. A pooled AUROC averages $\mathbb{1}[s_w > s_r]$ (ties $\frac{1}{2}$) over all wrong–right pairs. Partitioning the $K(QN-K)$ pairs into the $\sum_x k_x(N-k_x)$ same-question pairs and the rest gives the stated convex combination. For the expectation, use $\mathbb{E}[k_x(N-k_x)] = N(N-1)\,\mathbb{E}[p(1-p)]$ and $K \approx QN\bar{p}$.

Theorem 1. *Repair.* A fresh draw for question x is right with probability p_x, so replacing attempt i changes expected accuracy by $p_x - Z_i$, and a flag f_i contributes $f_i(p_x - Z_i)$. Since $\mathbb{E}[Z \mid x] = p_x$,

$$\mathbb{E}[f(p_x - Z) \mid x] = -\mathrm{Cov}(f, Z \mid x).$$

If f depends only on x, this covariance is 0.

The optimal flag sets $f = 1$ exactly when $\mathbb{E}[p_x - Z \mid x, s] > 0$, that is, when $P(Z{=}1 \mid x, s) < p_x$.

Selection. Given the answers and the scores, the posterior that answer group C_a is right is proportional to $\pi(\text{answers} \mid a \text{ right}) \prod_{i \in C_a} f_1(s_i \mid x) \prod_{i \notin C_a} f_0(s_i \mid x)$. Dividing by $\prod_i f_0(s_i \mid x)$ leaves $\pi(\cdot) \prod_{i \in C_a} \Lambda_x(s_i)$. If $f_1 = f_0$ given x, then $\Lambda \equiv 1$ and only the vote pattern remains. □

Proposition 1. Majority vote and the lexicographic rule differ only in tied configurations.

- *Four singletons, one of them right:* the rule is right iff the right attempt has the top score, with probability $M_4(d) = \int \phi(t-d)\Phi(t)^3 dt$, against $\frac{1}{4}$ for a random tie-break.
- *Two groups of two:* the rule compares group means, which differ by $\mathcal{N}(d, 1)$, so it is right with probability $\Phi(d)$, against $\frac{1}{2}$.

Both increments are non-negative for $d \geq 0$.

Overturn probability. Write the log-likelihood ratio of a score as $\ell(s) = ds - d^2/2$. Then ℓ is $\mathcal{N}(d^2/2, d^2)$ for a right attempt and $\mathcal{N}(-d^2/2, d^2)$ for a wrong one. For a right group of size k against a wrong group of size m, the score difference is

$$(k-m)\beta + \textstyle\sum_C \ell - \sum_W \ell$$
$$\sim \mathcal{N}\big((k-m)\beta + (k+m)d^2/2,\ (k+m)d^2\big).$$

Proposition 2. By the law of large numbers, $\frac{1}{N}\sum_{i \in C_a} \ell_i \to \pi(a)\,\mathbb{E}[\ell \mid Z_a]$, with $\mathbb{E}[\ell \mid Z{=}1] = \mathrm{KL}(f_1 \| f_0)$ and $\mathbb{E}[\ell \mid Z{=}0] = -\mathrm{KL}(f_0 \| f_1)$.

With group-shared noise of correlation ρ_s, a group mean keeps variance $\rho_s + (1-\rho_s)/n \to \rho_s$, which gives the stated cap.

For argmax with scores quantised to V verdicts, the top bin is reached with positive probability by both right and wrong attempts. The rule therefore converges to $P(Z{=}1 \mid \text{top bin})$.

Proposition 3. Within a question z_x is constant, so a right and a wrong attempt differ in score by $\theta_w d$ plus noise of variance $2(\theta_b^2 + \theta_w^2)$.

Under per-question balance, x_b is independent of Z given x, and also marginally, since $P(Z \mid x) = \frac{1}{2}$. The pointwise-optimal weight on it is therefore zero. At initialisation, the logistic gradient on any feature $\varphi(x)$ is $\mathbb{E}[(\sigma(0) - Z)\varphi(x)] = \mathbb{E}[(\frac{1}{2} - \frac{1}{2})\varphi(x)] = 0$.

Length law. Suppose step scores are Gumbel and m of T steps are shifted by $\log L$. The maximum of independent Gumbels with locations μ_i is Gumbel with location $\log \sum_i e^{\mu_i}$, and two independent Gumbels compare logistically. Hence the solution-level AUROC of a max-pooled detector is

$$\frac{T - m + mL}{2T - m + mL}.$$

A step-level AUROC of 0.80 ($L = 4$) gives 0.51–0.53 at $T = 100$–200, the level of the hidden-state error feature on own attempts in Phase 2.

B Robustness of CAV in the model

C Experimental details

Question sets.

- *test*: the 400-question FINAL-v3 SOLVE test split.
- *dev*: the 200-question development slice used in Phase 2, used here only for tuning.

Scenario	$d = 0.5$	$d = 1.0$
Baseline (ρ constant, $\rho_s = 0.3$)	+2.4	+5.3
Judge weaker on hard questions	+1.8	+4.2
Shared judge errors, $\rho_s = 0.6$	+1.7	+4.1
Variable attractor strength	+1.8 to +2.2	+3.1 to +4.1

Table 7: CAV ($N_1 = 16$, $m = 8$, probability scoring) minus majority vote at equal average cost (≈ 11 units), in points, in the calibrated model.

- *extra*: 400 SOLVE training questions that no run of the mixed adapter could have seen (all 1,134 SOLVE questions of its GRPO manifest and the trial slices are excluded).
- *train*: 450 further such questions for Stage 2.

Sets are fixed by a seeded script before any generation.

Sampling. Attempts use the SOLVE prompt with temperature 1.0, no top-p/top-k truncation and a 16,384-token cap. An attempt is correct if it finishes, passes the strict format gates and its last final-answer tag matches the key under the project's rule-based checker. Answers are grouped by checker equivalence.

Judging. Each answered attempt is split into steps: paragraphs, with display maths and very short paragraphs merged into the previous step, plus a final step "The final answer is X". The judge prompt is the v2 JUDGE prompt.

- A reasoning chain (temperature 1.0, 8,192-token cap) stops at "Decision:".
- The verdict probability is read from the top-20 next-token log-probabilities after appending "Decision:", taking the first of up to four forced positions that shows a verdict token. Chains that were cut off or never reached the label get a probability too.
- The *direct* probability appends "Decision:" right after the prompt, with no reasoning.
- Test attempts 0–3 get four chains each; all other attempts one.

Costs and statistics. Unit costs (pre-registered): attempt $= 1$, chain $= 0.3$, direct read $= 0.05$ (added after the smoke test). Decoded-token costs count attempt and chain tokens and exclude prefill. Every rule is averaged over 100 random orders of each question's 16 attempts. Majority at a fractional cost mixes the two neighbouring N. 95% intervals use 1,000–2,000 bootstrap resamples of questions; within-question AUROCs are bootstrapped over questions (cluster bootstrap).

Pre-registration.
`phase3/PREREGISTRATION.md` fixed the sets, sampling, judging, primary rule, primary comparison and success criterion before any attempt was generated. It also records three deviations and the semi-synthetic gain predictions made before any judging, each with its hash.